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Section : BAI-2A

LabTask : 12

Question 1

Code:

```
#include<iostream>
using namespace std;

class PaymentSystem
{
protected:
    int OriginalAmount;
    float serviceFee;
    float TotalAmount;
    float RefundAmount;

public :
    virtual void pay() = 0;
    virtual void refund() = 0;

    virtual void display()
    {
        cout<<"Original amount : "<<OriginalAmount<<endl;
        cout<<"service Fee : "<<serviceFee<<endl;
        cout<<"Total amount : "<<TotalAmount<<endl;
        cout<<"Refund Amount : "<<RefundAmount<<endl;
    }
};

class EasyPaisa : public PaymentSystem
{
public :

    void pay() override
    {
        cout<<"Enter the payment : ";
        cin>>OriginalAmount;
    }
}
```

```
void servicefee()
{
    serviceFee = OriginalAmount * 0.02;
}

void totalamount()
{
    TotalAmount = OriginalAmount - serviceFee;
}

void refund() override
{
    RefundAmount = OriginalAmount - serviceFee;
}

void display()
{
    cout << "\n--- EasyPaisa Receipt ---\n";
    servicefee();
    totalamount();
    refund();
    PaymentSystem::display();
}

};
```

```
class JazzCash: public PaymentSystem
{
public :

void pay() override
{
    cout<<"Enter the payment : ";
    cin>>OriginalAmount;
}

void servicefee()
{
    serviceFee = OriginalAmount * 0.05;
}

void totalamount()
{
    TotalAmount = OriginalAmount - serviceFee;
}

void refund() override
{
    RefundAmount = OriginalAmount - serviceFee;
}

void display()
{
    cout << "\n--- JazzCash Receipt ---\n";
    servicefee();
    totalamount();
    refund();
    PaymentSystem::display();
}
};
```

```
int main()
{
    PaymentSystem * ptr;
    JazzCash J1;
    EasyPaisa E1;

    cout<<"++++++ Jazz Cash Transaction ++++++ "<<endl;
    ptr = &J1;
    ptr->pay();
    ptr->display();

    cout<<"\n++++++ Easy Paisa Transaction ++++++ "<<endl;
    ptr = &E1;
    ptr->pay();
    ptr->display();

    return 0;
}
```

Ouput:

```
+++++++ Jazz Cash Transaction ++++++  
Enter the payment : 1000
```

```
--- JazzCash Receipt ---  
Original amount : 1000  
service Fee : 50  
Total amount : 950  
Refund Amount : 950
```

```
+++++++ Easy Paisa Transaction ++++++  
Enter the payment : 1000
```

```
--- EasyPaisa Receipt ---  
Original amount : 1000  
service Fee : 20  
Total amount : 980  
Refund Amount : 980
```

Question 2

Code:

```
#include<iostream>
using namespace std;

class FeeSystem
{
    protected :
    int TransactionFee = 100;
    int SubmittedAmount;
    int FinalAmount;
    int RefundAmount;

    public :
    virtual void submit_fee() = 0;
    virtual void return_fee() = 0;

    virtual void display()
    {
        cout<<"Submitted Fee : "<<SubmittedAmount<<endl;
        cout<<"Total paid : "<<FinalAmount<<endl;
        cout<<"refund Details : "<<RefundAmount<<endl;
    }
};
```

```
class BankTransfer : public FeeSystem
{
    public :

    void submit_fee()
    {
        cout<<"Enter the amount for Bank Transfer : ";
        cin>>SubmittedAmount;
    }

    void return_fee()
    {
        RefundAmount = FinalAmount;
    }

    void totalAmount()
    {
        FinalAmount = SubmittedAmount - TransactionFee;
    }

    void display()
    {
        totalAmount();
        return_fee();
        FeeSystem::display();
    }
};
```

```
int main()
{
    FeeSystem *ptr ;
    BankTransfer B1;
    CreditCard C1;

    ptr = &B1;
    ptr->submit_fee();
    ptr->display();

    ptr = &C1;
    ptr->submit_fee();
    ptr->display();

    return 0;
}
```

Ouput:

Enter the amount for Bank Transfer : 4000

Submitted Fee : 4000

Total paid : 3900

refund Details : 3900

Enter the amount for credit Card : 6000

Submitted Fee : 6000

Total paid : 6180

refund Details : 6000